# Power of a point and cyclic quadrilaterals

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L3-GE1 Level 3 Senior · Geometry

Turn circle diagrams into equations using equal tangents, the power of a point and the tests for four points being concyclic.
**By the end you can** use equal tangent lengths and the power of a point to turn a circle diagram into equations; prove that four points are concyclic, including with the converse of power of a point; combine power of a point with Pythagoras and similar triangles to find lengths at SMC, AMC 12 and early AIME level.
**Prerequisites:** Angle chasing and polygons (L2-GE1), Area, Pythagoras and similar shapes (L2-GE2).
**Contests:** UKMT SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 90 minutes.

## 1 Lesson

Two chords $AB$ and $CD$ of a circle cross at $P$, with $AP = 4$, $PB = 6$ and $CP = 3$. Neither the radius nor any angle is given, but $PD$ is determined, with $PD = 8$. This is a special case of the *power of a point*, a single theorem which also covers tangents, lines through a point outside the circle, and many questions of the form "is this quadrilateral cyclic?". At SMC and AMC 12 level, many circle questions that seem to need trigonometry can be done in two or three lines with this theorem.

### 1.1 Tangent lengths

**Equal tangents.**
The two tangents from an external point $P$ to a circle have equal length, and the line from $P$ to the centre bisects the angle between them.

*Proof.* Let the tangents touch the circle at $S$ and $T$, and let $O$ be the centre. A tangent is perpendicular to the radius at the point of contact, so $\angle OSP = \angle OTP = 90°$. Also $OS = OT$ and $OP$ is shared, so triangles $OSP$ and $OTP$ are congruent (RHS). Hence $PS = PT$ and $\angle SPO = \angle TPO$.

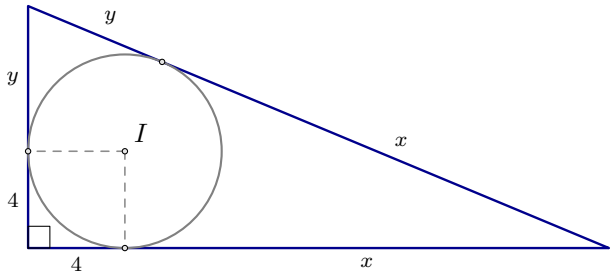
The most common application is to the incircle. If the incircle of triangle $ABC$ touches $BC$, $CA$, $AB$ at $D$, $E$, $F$, the six tangent segments come in three equal pairs, with $AE = AF = x$, $BF = BD = y$ and $CD = CE = z$. Adding all six gives $2(x + y + z) = a + b + c$, so $x + y + z = s$, the semiperimeter, and therefore

$$AE = AF = s - a, \qquad BF = BD = s - b, \qquad CD = CE = s - c,$$

where $a = BC$, $b = CA$ and $c = AB$. In the same way, if a circle touches all four sides of a quadrilateral $ABCD$, then $AB + CD = BC + DA$, because each side of that equation contains each of the four tangent lengths exactly once.



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**Example 1.**
A right-angled triangle has an incircle of radius 4 and a hypotenuse of length 26. Find its perimeter.

*Solution.*
Calling the legs $a$ and $b$ gives $a^2 + b^2 = 676$, which is one equation in two unknowns. The inradius has not been used yet, and tangent lengths are how the inradius enters.
Note that at the right angle, the incentre and the two points of contact on the legs form a square of side 4, since both radii are perpendicular to the legs. So the two tangents from the right-angled vertex have length 4. Call the tangent lengths from the other two vertices $x$ and $y$. The hypotenuse is made of one tangent from each of those vertices, so $x + y = 26$. The perimeter consists of every tangent length twice, so the perimeter equals

$$2(4 + x + y) = 2(4 + 26) = 60.$$

The legs were not needed. For the record, $a + b = 34$ and $a^2 + b^2 = 676$ give $ab = 240$, so the triangle is 10, 24, 26, and indeed $\frac{1}{2}(10 + 24 - 26) = 4$.

## 1.2 The power of a point

**Power of a point.**
Let $\omega$ be a circle with centre $O$ and radius $r$, and let $P$ be any point. If a line through $P$ meets $\omega$ at $A$ and $B$, then

$$PA \cdot PB = \left|OP^2 - r^2\right|,$$

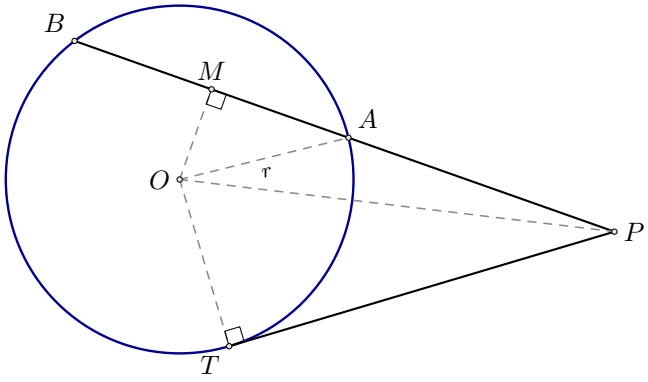
whichever line through $P$ you choose. In particular:

- if chords $AB$ and $CD$ meet at a point $P$ inside the circle, then $PA \cdot PB = PC \cdot PD$.
- if two lines through an outside point $P$ meet the circle at $A, B$ and at $C, D$, then $PA \cdot PB = PC \cdot PD$.
- if $P$ is outside and $PT$ is a tangent, then $PT^2 = PA \cdot PB$.

The number $OP^2 - r^2$ is called the power of $P$. The power is negative inside the circle, zero on the circle and positive outside.



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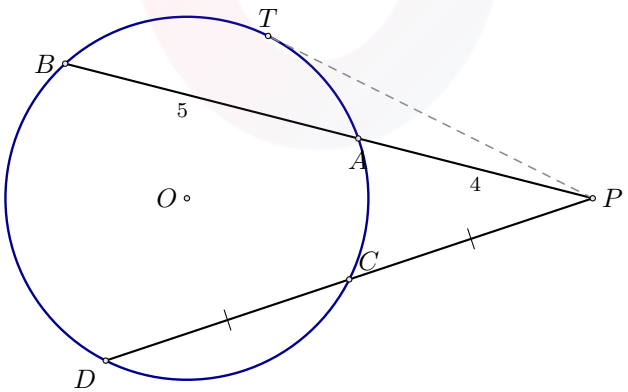



*Proof.* Drop the perpendicular from $O$ to the line, meeting the line at $M$. The perpendicular from the centre to a chord bisects that chord, so $MA = MB$. Suppose first that $P$ is outside, as in the diagram, and label so that $A$ is the nearer point. Then $PA = PM - MA$ and $PB = PM + MA$, so

$$PA \cdot PB = PM^2 - MA^2 = (OP^2 - OM^2) - (r^2 - OM^2) = OP^2 - r^2,$$

using Pythagoras in the right-angled triangles $OMP$ and $OMA$. If $P$ is inside, then with $A$ again the nearer point, $PA = MA - PM$ and $PB = MA + PM$, and the same calculation gives $r^2 - OP^2$. For the tangent, Pythagoras in triangle $OTP$ gives $PT^2 = OP^2 - r^2$ directly. The answer involves only $OP$ and $r$, so every line through $P$ gives the same product.

I include a second proof by similar triangles, because the same triangles turn up in harder problems. For chords $AB$ and $CD$ meeting at $P$, we have $\angle PAC = \angle BAC = \angle BDC = \angle PDB$ (angles in the same segment) and $\angle APC = \angle DPB$ (vertically opposite), so triangles $PAC$ and $PDB$ are similar and $PA/PD = PC/PB$. For a tangent, the alternate segment theorem gives $\angle PTA = \angle PBT$, so triangles $PTA$ and $PBT$ are similar and $PT/PA = PB/PT$.




**Example 2.**
Point $P$ lies outside a circle. One line through $P$ meets the circle at $A$ and $B$, with $PA = 4$ and $AB = 5$. A second line through $P$ meets the circle at $C$ and $D$, with $C$ between $P$ and $D$ and $PC = CD$. Find $PC$, and the length of a tangent from $P$ to the circle.

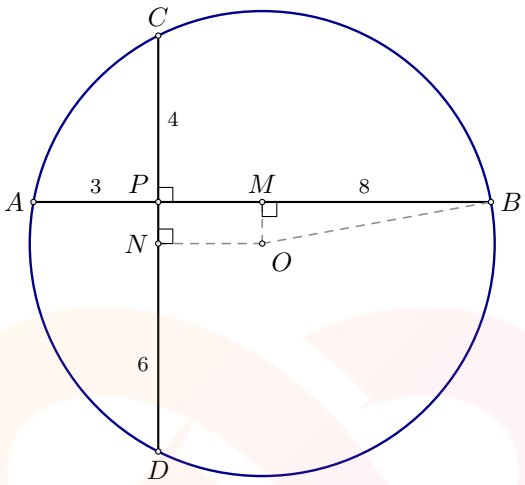
*Solution.*
A common slip is to multiply $PA$ by $AB$. Power of a point always measures both lengths from $P$. Here $PB = PA + AB = 9$, so the power of $P$ is $PA \cdot PB = 36$.



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On the second line $PD = 2\,PC$, so $PC \cdot 2PC = 36$, giving $PC^2 = 18$ and $PC = 3\sqrt{2}$. A tangent $PT$ satisfies $PT^2 = 36$, so $PT = 6$.

Power of a point gives products of lengths. To get at the radius you usually need the centre as well, and the centre is easy to locate, since the centre lies on the perpendicular bisector of every chord.




**Example 3.**
Chords $AB$ and $CD$ of a circle are perpendicular and meet at $P$, with $AP = 3$, $PB = 8$ and $CP = 4$. Find the radius of the circle.

*Solution.*
First, power of a point gives the missing length. We get $PD = \frac{AP \cdot PB}{CP} = \frac{24}{4} = 6$.
To find the radius we locate the centre. The perpendicular bisector of $AB$ is perpendicular to $AB$, so the bisector is parallel to $CD$, and the perpendicular bisector of $CD$ is parallel to $AB$. So the position of the centre $O$ relative to $P$ can be read off without any angles.
Let $M$ and $N$ be the midpoints of $AB$ and $CD$. Since $AB = 11$, $PM = \frac{11}{2} - 3 = \frac{5}{2}$, and since $CD = 10$, $PN = 5 - 4 = 1$. As $OM \perp AB$ and $ON \perp CD$, the quadrilateral $PMON$ is a rectangle, so $OM = PN = 1$. In the right-angled triangle $OMB$,

$$r^2 = OM^2 + MB^2 = 1^2 + \left(\tfrac{11}{2}\right)^2 = \tfrac{125}{4},$$

so $r = \frac{5\sqrt{5}}{2}$.
More generally, whenever perpendicular chords meet at $P$, we get $PA^2 + PB^2 + PC^2 + PD^2 = 4r^2$, and indeed $9 + 64 + 16 + 36 = 125$. Proving this is a good exercise. Put $P$ at the origin with the chords along the axes.

## 1.3 Cyclic quadrilaterals

**Concyclic criteria.**
Four points $A$, $B$, $C$, $D$ lie on a circle if any one of these holds:

1. $ABCD$ is a convex quadrilateral with a pair of opposite angles adding to $180°$.

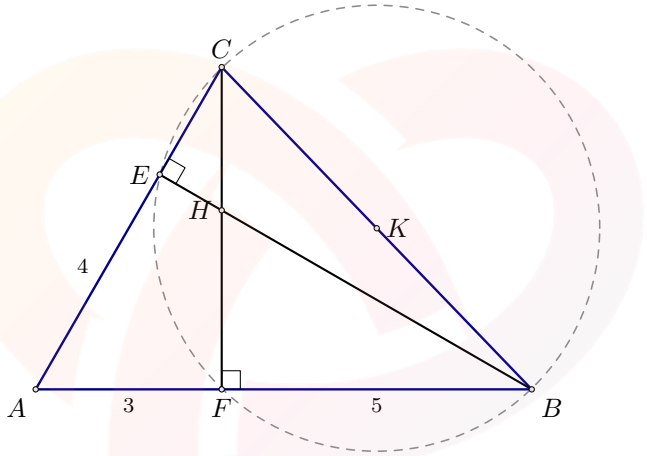


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2. $C$ and $D$ are on the same side of line $AB$ and $\angle ACB = \angle ADB$.
3. (converse of power of a point) lines $AB$ and $CD$ meet at $P$, the point $P$ lies either inside both segments $AB$, $CD$ or outside both, and $PA \cdot PB = PC \cdot PD$.

You met the first two in L2-GE1. The third is the converse of power of a point, and is often overlooked. Proof. Let the circle through $A$, $B$, $C$ meet line $PC$ again at $D'$. Power of a point gives $PA \cdot PB = PC \cdot PD'$, so $PD' = PD$. If $P$ is inside segment $AB$, then $P$ is inside the circle, so $D'$ is on the opposite side of $P$ from $C$, just as $D$ is. If $P$ is outside segment $AB$, then $P$ is outside the circle, so $D'$ and $D$ are both on the same side of $P$ as $C$. Either way $D' = D$.

There are two special cases you must know. If $\angle AXB = \angle AYB = 90°$, then $X$ and $Y$ lie on the circle with diameter $AB$. In particular, if $BE$ and $CF$ are altitudes of triangle $ABC$, then $B$, $C$, $E$, $F$ are concyclic, and the next example uses this.




**Example 4.**
In an acute triangle $ABC$ the altitudes $BE$ and $CF$ meet at $H$, with $E$ on $AC$ and $F$ on $AB$. Given $AF = 3$, $FB = 5$ and $AE = 4$, find $AC$ and $BC$.

*Solution.*
Pythagoras in triangle $AEB$ gives $BE^2 = 8^2 - 4^2 = 48$, but in triangle $AFC$ we know only $AF$, and nothing so far determines $C$. We need a relation involving $AC$, and finding that relation is the hard step.
Note that $\angle BEC = \angle BFC = 90°$, so $E$ and $F$ both lie on the circle with diameter $BC$ (dashed). Since angle $A$ is acute, $A$ lies outside that circle, and two lines through $A$ meet the circle, namely line $AB$ at $F$ and $B$, and line $AC$ at $E$ and $C$. Power of the point $A$ gives

$$AF \cdot AB = AE \cdot AC, \qquad 3 \cdot 8 = 4 \cdot AC,$$

so $AC = 6$ and $EC = 2$.
Pythagoras in triangle $BEC$ now gives $BC^2 = BE^2 + EC^2 = 48 + 4 = 52$, so $BC = 2\sqrt{13}$. As a check, $CF^2 = 6^2 - 3^2 = 27$ and $CF^2 + FB^2 = 27 + 25 = 52$ as well.
There is a second route. In right-angled triangle $AFC$, $\cos A = AF/AC$, and in triangle $AEB$, $\cos A = AE/AB$. Equating the two gives $AF \cdot AB = AE \cdot AC$ again.



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**Common traps.**

- Measure from $P$. For a line through $P$ meeting the circle at $A$ then $B$, the product is $PA \cdot PB$, never $PA \cdot AB$.
- Check the order of the points before writing $PB = PA + AB$. A diagram that is not to scale can put the nearer point on the wrong side.
- Do not assume points are concyclic because the picture looks round. Prove this with one of the criteria, namely right angles, equal angles on the same side, or the converse of power of a point.
- In a quadrilateral with an incircle, the equal sums are of *opposite* sides, namely $AB + CD = BC + DA$.
- A chord through an interior point $P$ is at least as long as the chord perpendicular to $OP$ and at most a diameter. If a question asks for all possible lengths, check both ends of that range.

### 1.4 How to spot this in a contest

For lengths around a circle, the strategies in short are:

1. If two lines pass through a common point and both meet the same circle, use power of a point, especially when the question gives lengths along those lines and no angles. A tangent from an external point is the case where the two intersection points coincide.
2. If no circle is drawn, look for one, because two right angles on the same segment, or two equal angles subtending the same segment on the same side, mean four points are concyclic whether or not the setter says so.
3. If the lengths must be integers (common in the AMC), factorise the power and check which factor pairs give a chord of possible length.
4. If a circle touches several sides of a polygon, label the tangent lengths before doing anything else, since the number of unknowns often drops from six to three or fewer.

## 2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** ⋆ Chords $AB$ and $CD$ of a circle meet at $P$. Given $AP = 6$, $PB = 4$ and $CD = 11$, find the length of the longer of the two segments $CP$ and $PD$.

**Problem 2** ⋆⋆ Point $P$ is at distance 13 from the centre of a circle of radius 5. A line through $P$ meets the circle at $A$ and $B$, with $A$ between $P$ and $B$, and $PA = 2\,AB$. Find $AB$.

**Problem 3** ⋆⋆ In a convex quadrilateral $ABCD$ the diagonals meet at $P$, with $AP = 4$, $PC = 9$, $BP = 6$ and $PD = 6$. Given that $AB = 5$, find $CD$.

**Problem 4** ⋆⋆⋆ A circle has centre $O$ and radius 7, and $P$ is a point with $OP = 3$. A chord $AB$



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passes through $P$, and both $PA$ and $PB$ are whole numbers. Find the sum of all possible lengths of $AB$.

**Problem 5** ⋆⋆⋆ Triangle $ABC$ has $AB = 13$, $BC = 14$ and $CA = 15$. The circle through $A$ that touches the line $BC$ at $B$ meets $CA$ again at $D$. Find $BD$.

**Problem 6** ⋆⋆⋆⋆ Triangle $ABC$ has $AB = 13$, $BC = 14$ and $CA = 15$. Its incircle touches $BC$ at $D$, and the segment $AD$ meets the incircle again at $X$. Find $\frac{XD}{AX}$.

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT SMC 2020 Question 15** ⋆
   *Two tangents from one point, so split the region along the line of symmetry*
2. **UKMT SMC 2015 Question 12** ⋆⋆
   *Equal tangents make isosceles triangles, so chase angles*
3. **UKMT SMC 2016 Question 24** ⋆⋆
   *Label the tangent lengths from each external point, then Pythagoras*
4. **AMC 12A 2013 Problem 19** ⋆⋆⋆
   *The power of C with respect to the circle, then a size check*
5. **UKMT SMC 2017 Question 19** ⋆⋆⋆
   *The corner of the rectangle has a known power, or drop a perpendicular from the centre*
6. **AMC 12B 2019 Problem 20** ⋆⋆⋆
   *Equal tangents put the meeting point on the perpendicular bisector of the chord*
7. **AIME 1999 Problem 12** ⋆⋆⋆
   *Tangent lengths give every side in terms of one unknown, so compare two area formulas*
8. **UKMT Maclaurin 2022 Question M6** ⋆⋆⋆
   *Angles in the same segment give a similar pair sharing the angle at C*
9. **AIME I 2024 Problem 10** ⋆⋆⋆⋆
   *Tangent-chord angles, then the power of the point where the tangents meet*
10. **UKMT Maclaurin 2021 Question M3** ⋆⋆⋆⋆
    *Write down the power of A with respect to each circle*
11. **AIME I 2019 Problem 13** ⋆⋆⋆⋆
    *Find a cyclic angle equality, then meet line CF with line AB and use power of a point*
12. **AIME I 2016 Problem 15** ⋆⋆⋆⋆
    *The line XY bisects the common tangent, so compute powers along the line*



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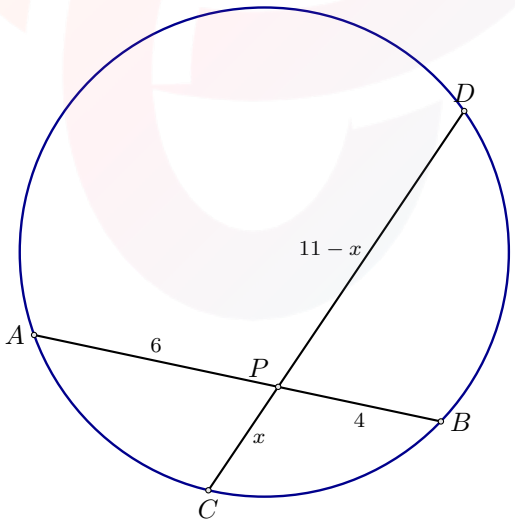
## 4 Hints

1. Write $CP = x$ and $PD = 11 - x$. You already know the product $CP \cdot PD$.
2. Find the power of $P$ from the distance to the centre and the radius, then write everything in terms of $AB$.
3. Compare $PA \cdot PC$ with $PB \cdot PD$. What does that tell you about the four vertices?
4. Power of a point gives the product $PA \cdot PB$. Then ask what the shortest and longest chords through $P$ are.
5. Look at the point $C$, since one of the lines through $C$ is a tangent. The similar triangles behind the tangent case give more than the power alone.
6. The tangent length from $A$ to the incircle is easy to find, and then power of a point needs only one more length.

## 5 Solutions

### Problem 1

*Chords $AB$ and $CD$ of a circle meet at $P$. Given $AP = 6$, $PB = 4$ and $CD = 11$, find the length of the longer of the two segments $CP$ and $PD$.*




By the intersecting chords theorem, $CP \cdot PD = AP \cdot PB = 24$. With $CP = x$ and $PD = 11 - x$ this reads $x(11 - x) = 24$, that is $x^2 - 11x + 24 = 0$, which factorises as $(x - 3)(x - 8) = 0$. So the two segments have lengths 3 and 8, and the longer one is 8.

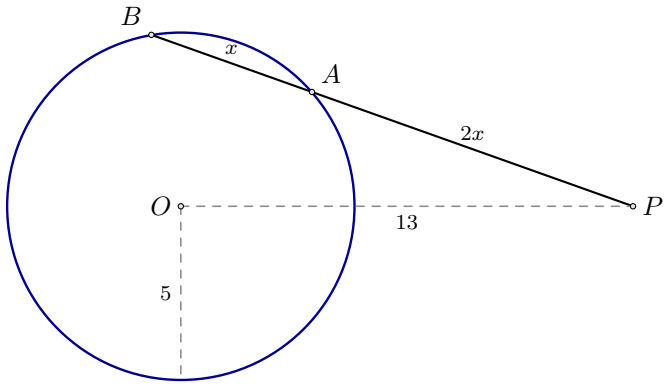
**Answer:** 8

### Problem 2

*Point $P$ is at distance 13 from the centre of a circle of radius 5. A line through $P$ meets the circle at $A$ and $B$, with $A$ between $P$ and $B$, and $PA = 2\,AB$. Find $AB$.*



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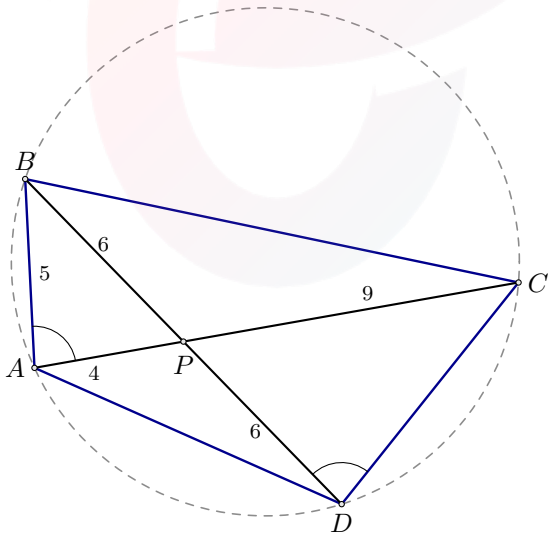

The power of $P$ is $OP^2 - r^2 = 169 - 25 = 144$, and equivalently a tangent from $P$ has length 12. Write $AB = x$. Then $PA = 2x$ and $PB = PA + AB = 3x$, so $PA \cdot PB = 6x^2 = 144$. This gives $x^2 = 24$, so $AB = 2\sqrt{6}$.

We should check that such a line exists. The nearer intersection point is at distance between $OP - r = 8$ (the line through the centre) and 12 (the tangent), and $PA = 4\sqrt{6} \approx 9.8$ lies in that range.

**Answer:** $2\sqrt{6}$

### Problem 3

*In a convex quadrilateral $ABCD$ the diagonals meet at $P$, with $AP = 4$, $PC = 9$, $BP = 6$ and $PD = 6$. Given that $AB = 5$, find $CD$.*




We have $PA \cdot PC = 36 = PB \cdot PD$, and $P$ lies inside both diagonals, so by the converse of the intersecting chords theorem $ABCD$ is cyclic. Then $\angle BAC = \angle BDC$ (angles in the same segment), that is $\angle PAB = \angle PDC$, and $\angle APB = \angle DPC$ (vertically opposite). So triangles $APB$ and $DPC$ are similar, with $A$, $P$, $B$ corresponding to $D$, $P$, $C$. The ratio of similarity is $DP/AP = 6/4 = \frac{3}{2}$, so $CD = \frac{3}{2} \cdot AB = \frac{15}{2}$.

As a check, $CP/BP$ should also equal $\frac{3}{2}$, and indeed $9/6 = \frac{3}{2}$. In fact that check is a proof on its own, since $AP/DP = BP/CP$ with the vertically opposite angles gives the similarity directly (SAS). So the converse of power of a point is equivalent to this SAS similarity.

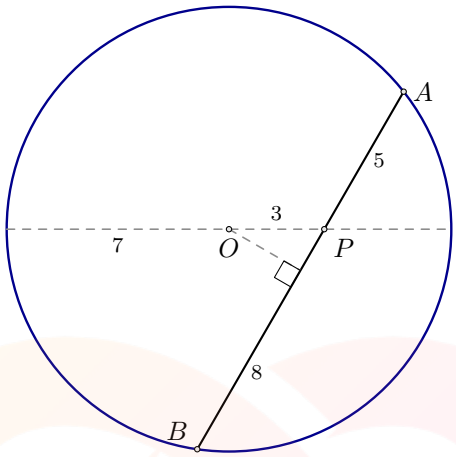


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**Answer:** $\frac{15}{2}$

### Problem 4

*A circle has centre $O$ and radius 7, and $P$ is a point with $OP = 3$. A chord $AB$ passes through $P$, and both $PA$ and $PB$ are whole numbers. Find the sum of all possible lengths of $AB$.*




The power of $P$ gives $PA \cdot PB = r^2 - OP^2 = 49 - 9 = 40$.

Next we find the possible lengths of a chord through $P$. A chord at distance $d$ from $O$ has length $2\sqrt{49 - d^2}$, and a chord through $P$ has $0 \le d \le OP = 3$, with every such $d$ occurring as the chord turns about $P$. So the possible lengths are exactly those from $2\sqrt{40} \approx 12.6$ (the chord perpendicular to $OP$) up to 14 (the diameter).

The factor pairs of 40 are $1 \times 40$, $2 \times 20$, $4 \times 10$ and $5 \times 8$, with sums 41, 22, 14 and 13. Only 14 and 13 lie in the allowed range. The sum 14 is the diameter through $P$, with $PA = 7 - 3 = 4$ and $PB = 7 + 3 = 10$, and the sum 13 is a chord with $PA = 5$ and $PB = 8$. The answer is $13 + 14 = 27$. Forgetting the diameter is the trap here.

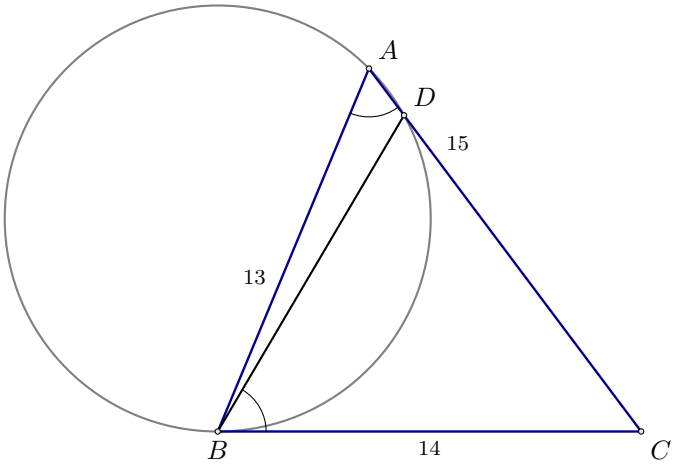
**Answer:** 27

### Problem 5

*Triangle $ABC$ has $AB = 13$, $BC = 14$ and $CA = 15$. The circle through $A$ that touches the line $BC$ at $B$ meets $CA$ again at $D$. Find $BD$.*



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From $C$, the line $CB$ is tangent at $B$ and the line $CA$ meets the circle at $D$ and $A$, so power of a point gives $CB^2 = CD \cdot CA$, that is $CD = \frac{196}{15}$. This is less than 15, so $D$ lies between $A$ and $C$.

For $BD$, use the similar triangles behind the tangent case. The line $BD$ separates $A$ from $C$, so by the alternate segment theorem the angle between the tangent $BC$ and the chord $BD$ equals the angle that $BD$ subtends at $A$, so $\angle CBD = \angle BAD = \angle BAC$. Triangles $CBD$ and $CAB$ also share the angle at $C$, so they are similar, with $C$, $B$, $D$ corresponding to $C$, $A$, $B$. Hence $BD/AB = CB/CA$, and

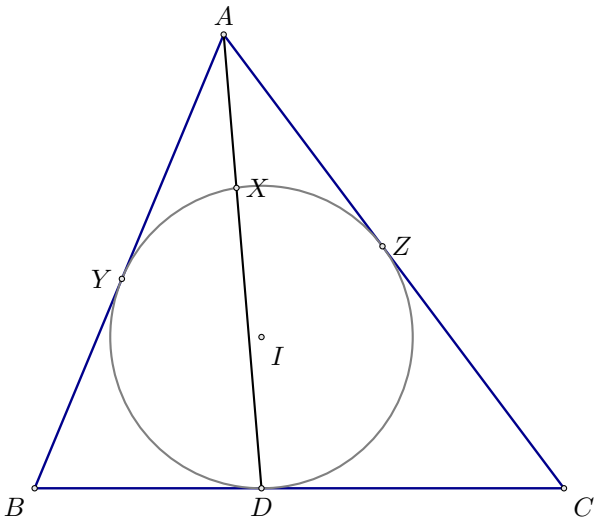
$$BD = 13 \cdot \frac{14}{15} = \frac{182}{15}.$$

The same similarity gives $CD/CB = CB/CA$, which is the power of a point statement again.

**Answer:** $\frac{182}{15}$

### Problem 6

*Triangle $ABC$ has $AB = 13$, $BC = 14$ and $CA = 15$. Its incircle touches $BC$ at $D$, and the segment $AD$ meets the incircle again at $X$. Find $\frac{XD}{AX}$.*






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We start with tangent lengths. The semiperimeter is $s = 21$, so the tangents from $A$ to the incircle (to the points $Y$ and $Z$ in the diagram) have length $s - a = 21 - 14 = 7$, and $BD = s - b = 21 - 15 = 6$. Power of the point $A$ with respect to the incircle gives

$$AX \cdot AD = 7^2 = 49.$$

Finding $AD$ remains, which is fairly routine with coordinates. Put $B = (0,0)$ and $C = (14,0)$, and let $A = (x,y)$. From $x^2 + y^2 = 169$ and $(x-14)^2 + y^2 = 225$, subtracting gives $28x - 196 = -56$, so $x = 5$ and then $y = 12$. Since $D = (6,0)$, we get $AD^2 = 1^2 + 12^2 = 145$.

Therefore $AX = \frac{49}{\sqrt{145}}$ and $XD = AD - AX = \frac{145-49}{\sqrt{145}} = \frac{96}{\sqrt{145}}$, so

$$\frac{XD}{AX} = \frac{96}{49}.$$

In general the ratio is $\frac{AD^2 - t^2}{t^2}$, where $t = 7$ is the tangent length, so $\sqrt{145}$ need not appear at all.

**Answer:** $\frac{96}{49}$